From a question to an answer

What was not working
Finding a usable public dataset takes days. The labels are inconsistent, the sample counts are not what they seem, and once you find one you still cannot see what it says.

What locus does
Ask in plain English. It searches the public archives, reads the sample records to count cases and controls, and then runs the first analysis on your laptop.

1 **Ask**
A question, typed or spoken

2 **Find**
Ranked studies, with counted facts

3 **Analyse**
Counts, statistics and figures

Open source · Apache-2.0 · github.com/UoSDDNB/omics-agent

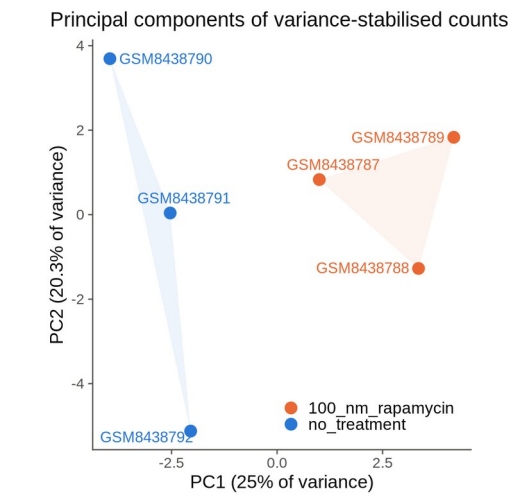
Live

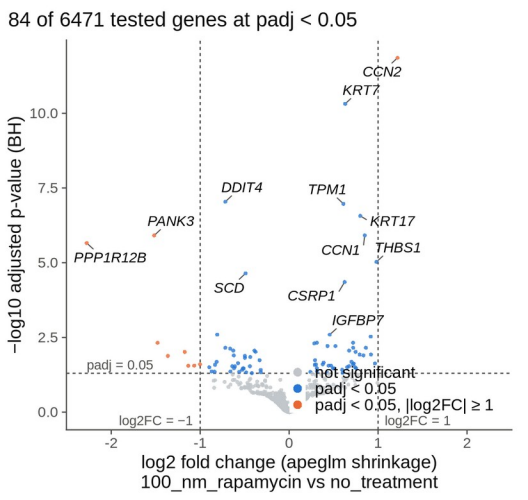
The question we are about to ask

> *“bulk RNA-seq of human cells treated with an mTOR inhibitor such as rapamycin, compared with vehicle”*

- Search the archives
- Open the study it ranks first, with its cases and controls already counted
- Press Process

Three minutes later, on a laptop





84

genes changed
(padj < 0.05)

DDIT4 ↓

a gene that needs mTOR,
falling as expected

Treated and untreated samples separate. Nobody wrote any code.

GSE273888 · HeLa cells, rapamycin vs untreated, 3 + 3 · first 2 million reads per sample · every tool version recorded in a manifest

What it does not do yet

- Reads are subsampled for speed, so the gene counts are smaller than a full run.
- One comparison per dataset, and we choose which column marks treatment.
- Bulk RNA-seq only so far. Ribo-seq and proteomics are designed, not built.
- Proteomics sample counts need an SDRF file, which only some projects have.
- The heavy pipelines stay on a cluster. We export the files and the command for Nextflow and MaxQuant rather than pretending a laptop can run them.
- This machine has no index built, so today's search queries GEO, PRIDE and CELLxGENE live rather than all 18.
- The written summary comes from a model, and a check rejects any number it did not get from us.

What is underneath

18

repositories covered by the index

891k

datasets, once the index is built

9

MCP tools, so an agent can use it too

21

tests, green from a clean clone

Counted, not guessed. Cases and controls are counted in people where the records allow it, and reported as unknown where they do not. The model chooses which field marks treatment; the arithmetic is code.

Runs on your own hardware. Search works against a local model, so no data leaves the machine and no API key is needed.